

Assignment: Vuvuzela removal from broadcast audio

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If you have watched the FIFA World Cup, chances are you have been tormented by the incessant drone of the Vuvuzela. A Vuvuzela is a long plastic horn capable of producing a loud monotone note that is very annoying and drowns out TV announcers. One way to reduce the Vuvuzela noise and improve clarity of commentary in broadcasts is by using frequency-selecting filtering that selectively suppresses particular frequency components generated by the Vuvuzela. However, bear in mind that there may be some overlap between the frequency components of the Vuvuzela and the frequency components in speech. This means that too much filtering may attenuate the voices of commentators, which is undesirable.

Data: You will be given two audio files. One is a sample audio recording contaminated by Vuvuzela noise (`noisy_audio.wav`). The other is a sample of Vuvuzela sound with no other audio (`vuvuzela_example.wav`).

Your task: You are required to design a filtering scheme in MATLAB to remove the Vuvuzela noise from `noisy_audio.wav` with the aim of recovering as much audio clarity as possible while, at the same time, keeping the overall delay below 0.2 seconds.

Submissions: You will make the following submissions via Moodle for this assignment:

1. Submit your code (as a single `.m` file)
2. Submit a brief report (at most four A4 pages) in `.pdf` format explaining your design choices (see below for details)
3. Submit a brief (at most 5 minutes) video where you describe what each part of your code does, and run your code, demonstrating how well it denoises the noisy audio.

These submissions due date will be available on Moodle.

Marking: The assignment will be marked out of 10. Five marks are allocated to how effectively your design denoises the given signal in your demonstration video.

Five marks are allocated to your report. The following is a guide to how marking will be carried out.

Demonstration video (5 marks).

Suggestions for putting your video together will be posted separately.

- 5/5 Clear explanation given as to why the system you designed meets the delay requirement. Speech clearly audible and minimally distorted. Vuvuzela sound almost completely suppressed.

- 4/5 Speech audible (possibly with some noticeable distortion) and vuvuzela sound minimal.
- 3/5 Useful filtering occurs but some speech moderately distorted and/or some quite noticeable vuvuzela sound present. System may or may not meet delay
- 2/5 Some filtering achieved but speech highly distorted and/or vuvuzela only slightly removed.
- 1/5 Task clearly attempted but little or no useful filtering achieved.
- 0/5 No demonstration submitted.

Report (5 marks), maximum length four A4 pages (including diagrams)

- 1 mark *How did you come up with your filter design specifications?* Describe (including plots or diagrams) of how you came up with your filter design specifications from the given signals and the delay requirement.
- 1 mark *What were your design specifications?* Briefly describe your desired pass- and stop-band edges, desired filter lengths, and any other specifications you wanted your filter(s) to achieve.
- 1 mark *Tell us about the filters you designed.* Plot the frequency response (magnitude and phase). State the order(s) of the filter(s) you actually obtained using filter design tools based on your specifications.
- 1 mark *Tradeoffs and limitations:* Briefly discuss the trade-offs you encountered in developing your design, and any limitations you see with the approach you have taken.
- 1 mark *Presentation:* following the presentation guidelines for lab reports, including:
 - labelling clearly all of your plots
 - Writing your explanations in complete, clear, sentences.

Partial credit will be given for the points above where students make a clear effort, but the resulting discussion/description misses significant points or shows significant errors in understanding.